

Module 2: Components Deep Dive

Assignment 2.1 – Creating Components Using CLI

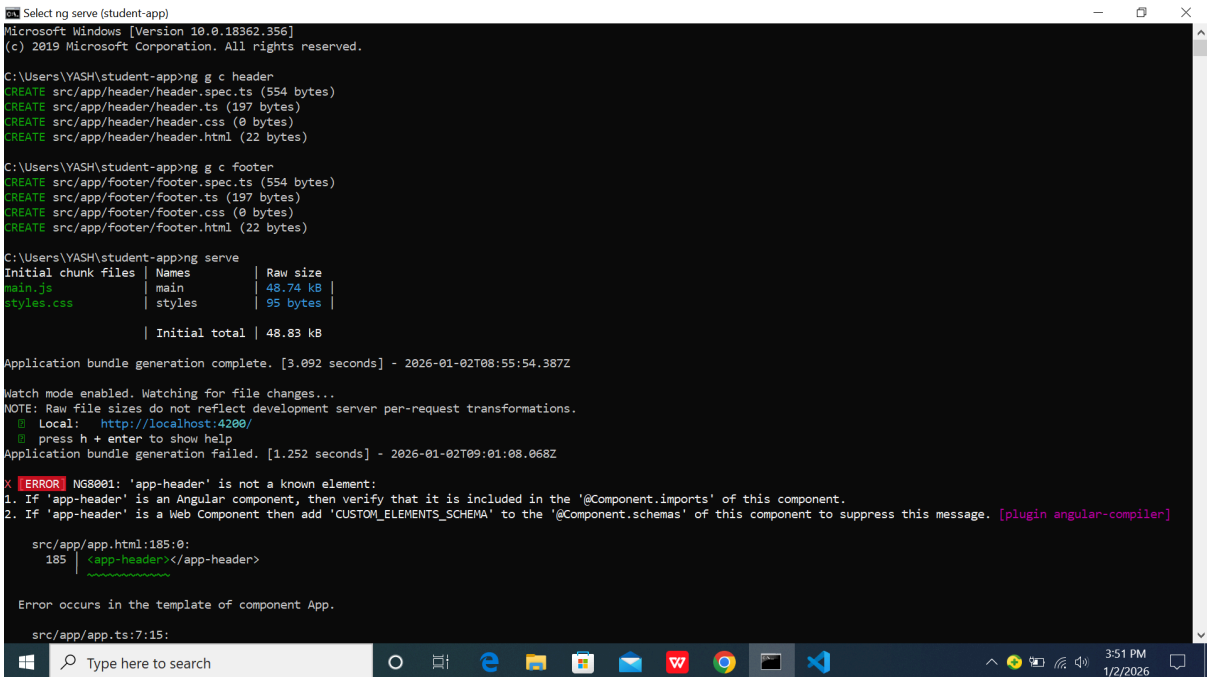
Objective:

Learn to create and use components.

Tasks:

1. Create the following components using Angular CLI:

- o header
- o footer



```
Microsoft Windows [Version 10.0.18362.356]
(c) 2019 Microsoft Corporation. All rights reserved.

C:\Users\YASH\student-app>ng g c header
CREATE src/app/header/header.spec.ts (554 bytes)
CREATE src/app/header/header.ts (197 bytes)
CREATE src/app/header/header.css (0 bytes)
CREATE src/app/header/header.html (22 bytes)

C:\Users\YASH\student-app>ng g c footer
CREATE src/app/footer/footer.spec.ts (554 bytes)
CREATE src/app/footer/footer.ts (197 bytes)
CREATE src/app/footer/footer.css (0 bytes)
CREATE src/app/footer/footer.html (22 bytes)

C:\Users\YASH\student-app>ng serve
Initial chunk files | Names | Raw size
main.js | main | 48.74 kB
styles.css | styles | 95 bytes
| Initial total | 48.83 kB

Application bundle generation complete. [3.092 seconds] - 2026-01-02T08:55:54.387Z

Watch mode enabled. Watching for file changes...
NOTE: Raw file sizes do not reflect development server per-request transformations.
@ Local: http://localhost:4200/
@ press h + enter to show help

Application bundle generation failed. [1.252 seconds] - 2026-01-02T09:01:08.068Z

X [ERROR] NG8001: 'app-header' is not a known element:
1. If 'app-header' is an Angular component, then verify that it is included in the '@Component.imports' of this component.
2. If 'app-header' is a Web Component then add 'CUSTOM_ELEMENTS_SCHEMA' to the '@Component.schemas' of this component to suppress this message. [plugin angular-compiler]

src/app/app.html:185:0:
185 | <app-header></app-header>
    | ~~~~~

Error occurs in the template of component App.

src/app/app.ts:7:15:
```

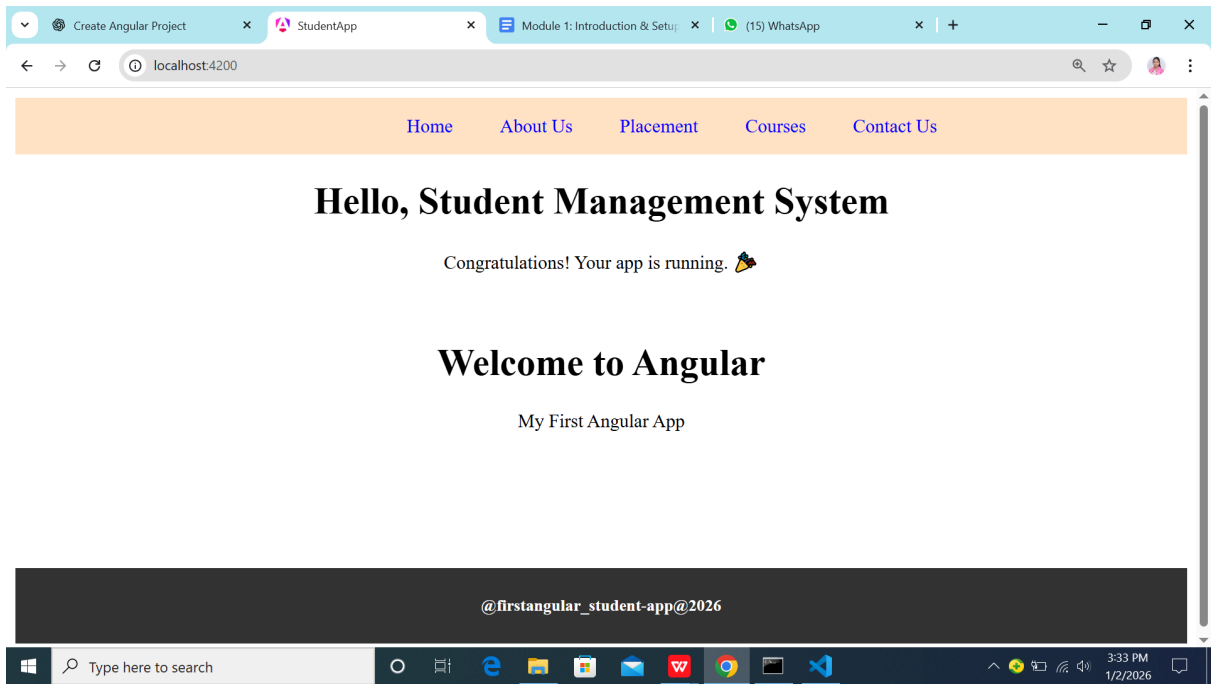
2. Display both components inside app.component.html.

Expected UI Layout:

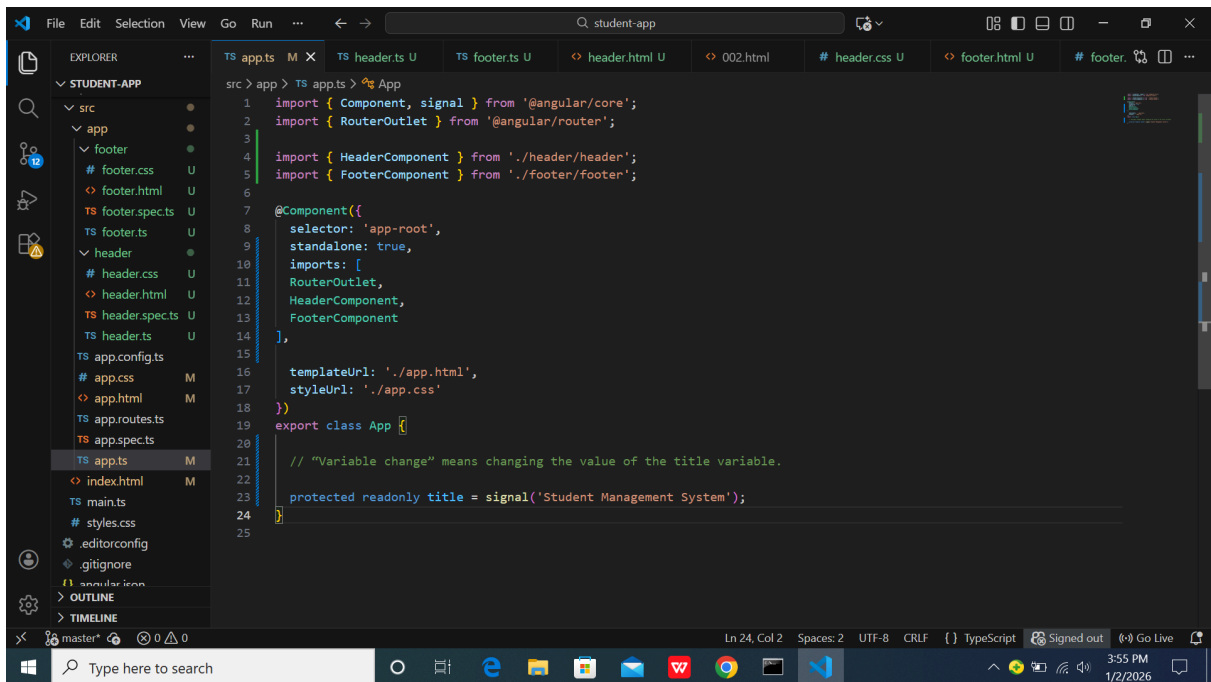
HEADER

Main Content

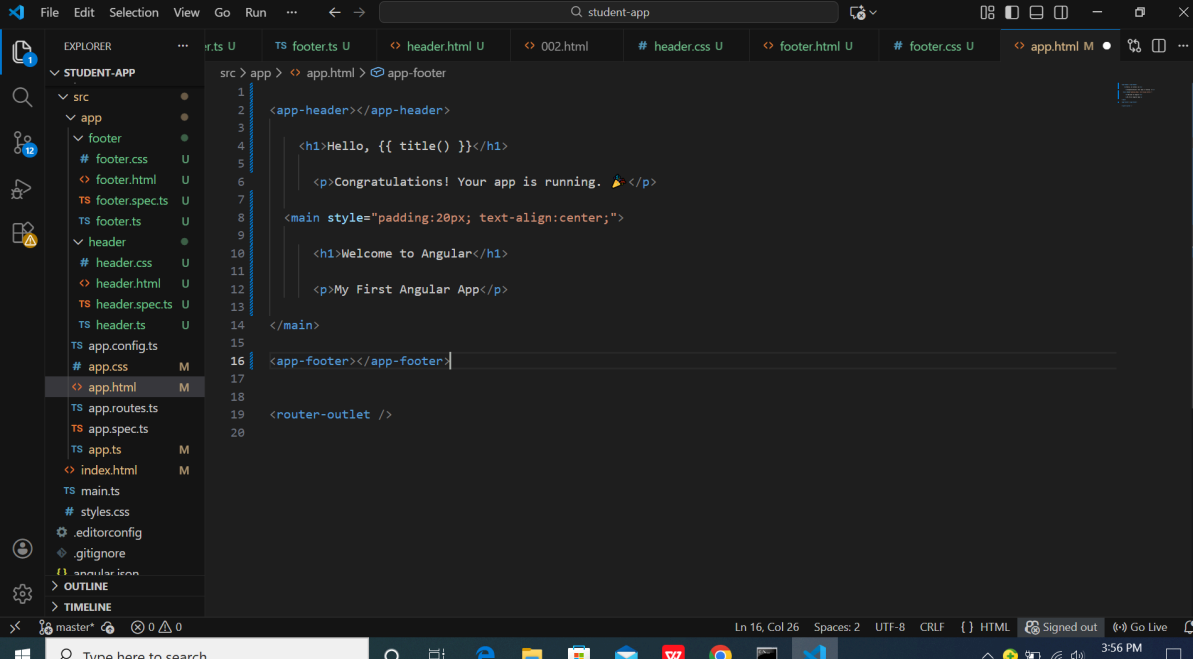
FOOTER



app.ts



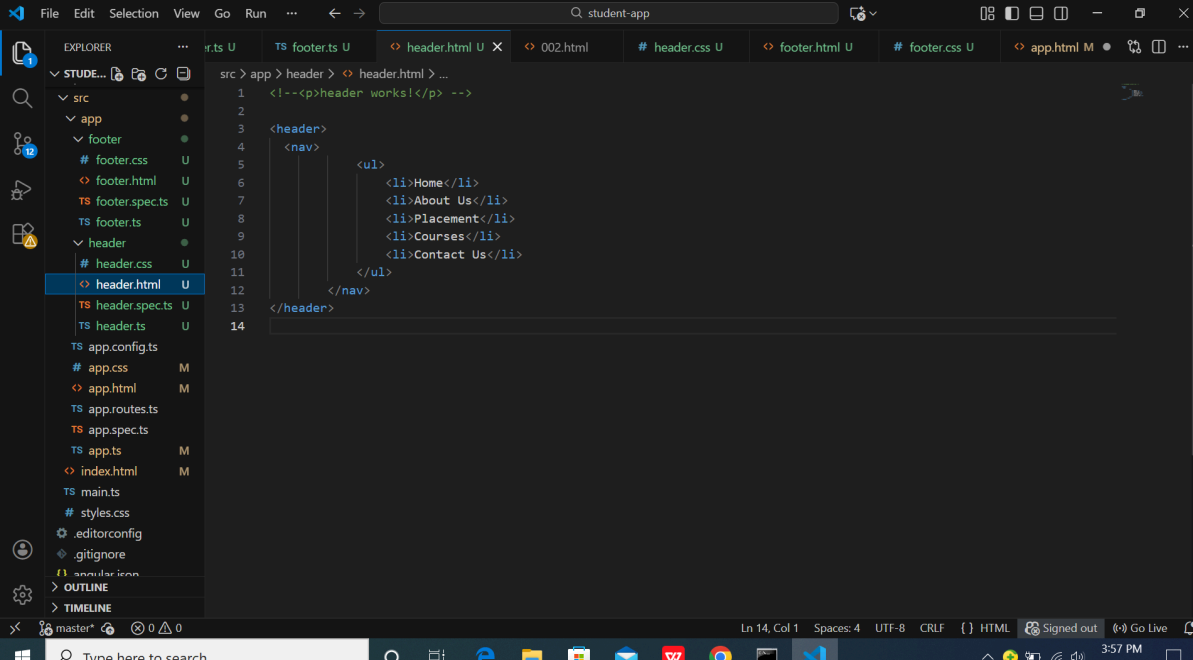
app.html



The screenshot shows the Visual Studio Code editor with the 'student-app' project open. The Explorer panel on the left shows the file structure, with 'app.html' selected under the 'app' directory. The main editor displays the content of 'app.html' with line numbers 1 through 20. The code includes an Angular component structure with a header, main content area, and footer. The footer contains a router-outlet.

```
1 <!--<p>header works!</p> -->
2
3 <header>
4   <nav>
5     <ul>
6       <li>Home</li>
7       <li>About Us</li>
8       <li>Placement</li>
9       <li>Courses</li>
10      <li>Contact Us</li>
11    </ul>
12  </nav>
13 </header>
14
15
16 <app-footer></app-footer>
17
18
19 <router-outlet />
20
```

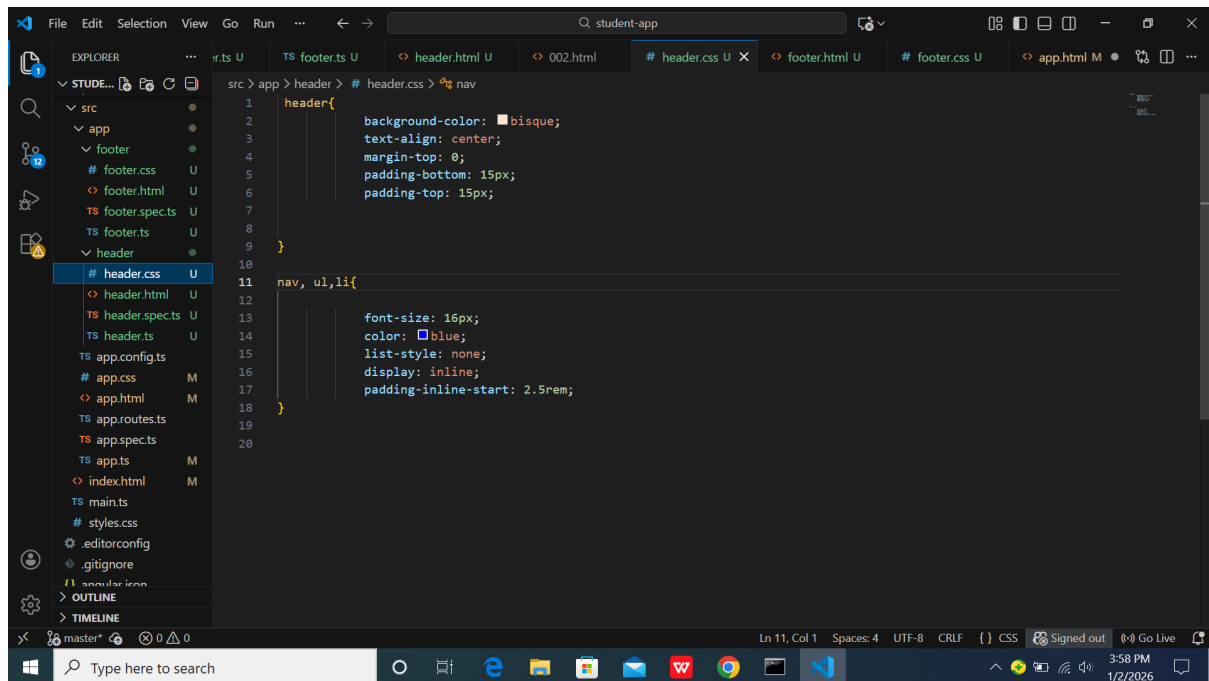
header.html



The screenshot shows the Visual Studio Code editor with the 'student-app' project open. The Explorer panel on the left shows the file structure, with 'header.html' selected under the 'header' directory. The main editor displays the content of 'header.html' with line numbers 1 through 14. The code includes an Angular component structure for a header with a navigation menu.

```
1 <!--<p>header works!</p> -->
2
3 <header>
4   <nav>
5     <ul>
6       <li>Home</li>
7       <li>About Us</li>
8       <li>Placement</li>
9       <li>Courses</li>
10      <li>Contact Us</li>
11    </ul>
12  </nav>
13 </header>
14
```

header.css

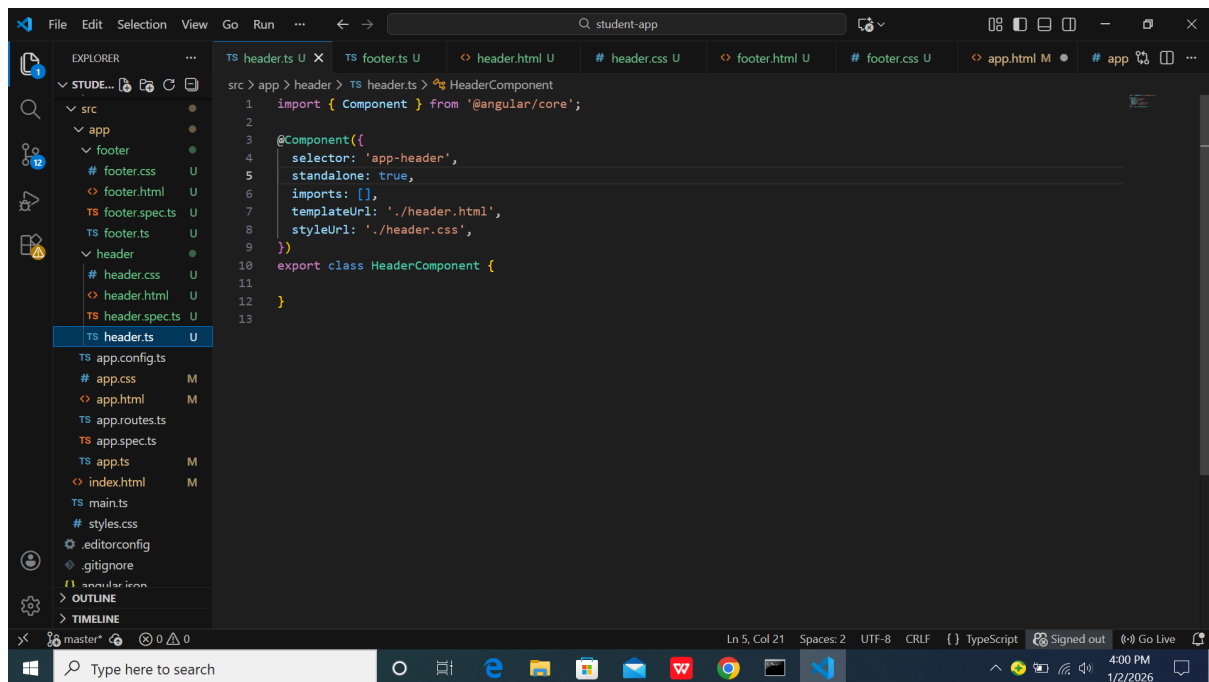


The screenshot shows the Visual Studio Code editor with the 'header.css' file open. The Explorer sidebar on the left shows the project structure with 'src' > 'app' > 'header' containing 'header.css', 'header.html', 'header.spec.ts', and 'header.ts'. The main editor area displays the following CSS code:

```
1 header{
2     background-color: #bisque;
3     text-align: center;
4     margin-top: 0;
5     padding-bottom: 15px;
6     padding-top: 15px;
7 }
8
9
10
11 nav, ul, li{
12
13     font-size: 16px;
14     color: blue;
15     list-style: none;
16     display: inline;
17     padding-inline-start: 2.5rem;
18 }
19
20
```

The status bar at the bottom indicates 'Ln 11, Col 1', 'Spaces: 4', 'UTF-8', 'CRLF', and 'CSS'.

header.ts



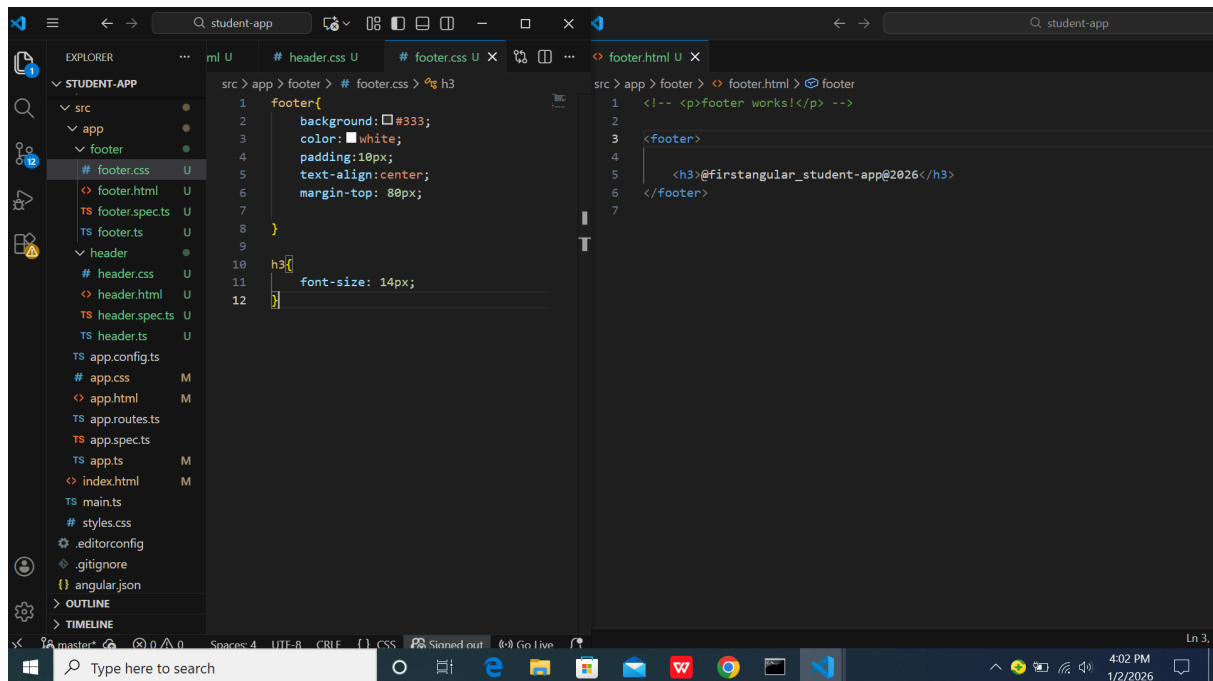
The screenshot shows the Visual Studio Code editor with the 'header.ts' file open. The Explorer sidebar on the left shows the project structure with 'src' > 'app' > 'header' containing 'header.css', 'header.html', 'header.spec.ts', and 'header.ts'. The main editor area displays the following TypeScript code:

```
1 import { Component } from '@angular/core';
2
3 @Component({
4     selector: 'app-header',
5     standalone: true,
6     imports: [],
7     templateUrl: './header.html',
8     styleUrls: ['./header.css'],
9 })
10 export class HeaderComponent {
11
12 }
13
```

The status bar at the bottom indicates 'Ln 5, Col 21', 'Spaces: 2', 'UTF-8', 'CRLF', and 'TypeScript'.

Footer.css

footer.html



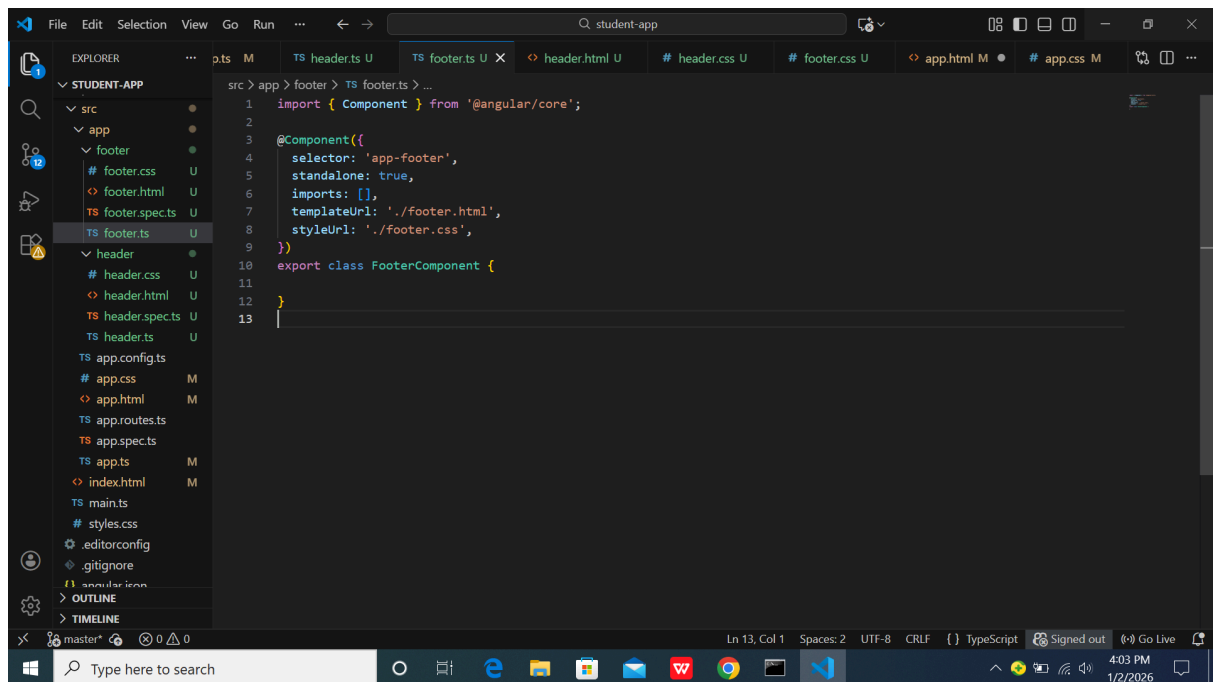
The screenshot shows the Visual Studio Code editor with two files open: `footer.css` and `footer.html`. The `EXPLORER` sidebar on the left shows the project structure for `STUDENT-APP`, including `src`, `app`, `footer`, `header`, and various configuration files. The `footer.css` file contains the following CSS rules:

```
1 footer{
2   background-color: #333;
3   color: white;
4   padding: 10px;
5   text-align: center;
6   margin-top: 80px;
7 }
8
9
10 h3{
11   font-size: 14px;
12 }
```

The `footer.html` file contains the following HTML structure:

```
1 <!-- <p>footer works!</p> -->
2
3 <footer>
4
5   <h3>@firstangular_student-app@2026</h3>
6 </footer>
7
```

footer.ts



The screenshot shows the Visual Studio Code editor with the `footer.ts` file open. The `EXPLORER` sidebar on the left shows the project structure for `STUDENT-APP`, including `src`, `app`, `footer`, `header`, and various configuration files. The `footer.ts` file contains the following TypeScript code:

```
1 import { Component } from '@angular/core';
2
3 @Component({
4   selector: 'app-footer',
5   standalone: true,
6   imports: [],
7   templateUrl: './footer.html',
8   styleUrls: ['./footer.css'],
9 })
10 export class FooterComponent {
11
12 }
13
```